

Zonglin Di

CONTACT INFORMATION	Tongji University, 1239, Siping Road Yangpu District, Shanghai, P. R. China, 201800	(+86) 131-2230-8670 1452640@tongji.edu.cn
RESEARCH INTERESTS	Automatic Road Extraction using deep learning, Imbalance Classification, Bio-Computation	
EDUCATION	Tongji University , Shanghai, China B.E., Control Science and Engineering, <i>Expected:</i> Summer 2019 • GPA: 4.25/5.0 • Related Courses: Higher Mathematics, Linear Algebra, Possibility Theory & Statistics, Analog Circuit, Digital Circuit, Signal & Systems, The Introduction to Pattern Recognition, Object-Oriented Programming, The Principle And Application to Database, Operating System, Compiler Principle, Digital Signal Processing, Digital Image Processing, Operating Research	
RESEARCH EXPERIENCE	Research Assistant of Deep Learning Laboratory led by Prof. Yin Wang • Extract the roadmap from satellite images and GPS data using deep learning methods including VGG-16, FCN, U-Net, DLink-Net, etc. in Keras, Tensorflow and PyTorch especially. • 1 research paper has been accepted (<i>GeoAI 2018</i>) and 1 conference paper of <i>CVPR 2019</i> is on the way. • Supervisor: Prof. Yin Wang, Doctor of University of Michigan. Research Intern in Intel Asia & Pacific R & D Oct 2017 to Mar 2018 • Warp Affine, SIFT, SURF using OpenCV3 & Python 2.7 • Implementation of FCN in TensorFlow and Keras • Implementation of VAE, C-VAE in TensorFlow and Keras • Directed by Mr. Jiajie Yao Research Assistant of State Lab. of Embedded System & Service Computing, Ministry of Education Dec 2014 to Present • Research on Internet Addiction Disorder using Machine Learning methods including clustering and classification • Research on EEG signal detection • 2 conference papers have been accepted (<i>ICSP-BMEI 2017</i>, <i>ICNC-FSKD 2016</i>) and 1 journal paper has been submitted • Supervisors: Prof. Asoke K. Nandi, <i>IEEE Fellow</i> Research Assistant of State Key Lab. of Management & Control for Complex Systems, Ministry of Education, Sept 2016 to Present • Research on imbalance classification using SVM, data distribution information and pre-clustering methods • 1 conference paper (<i>ICNSC 2018</i>) and 1 conference paper <i>SMC 2018</i> have been submitted • Supervisors: Prof. Mengchu Zhou, <i>IEEE Fellow</i> & Prof. Qi Kang, <i>IEEE Senior Fellow</i>	
PUBLICATIONS	1. Tao Sun, Zonglin Di and Yin Wang, “Combining Satellite Imagery and GPS Data for Road Extraction” <i>2nd ACM SIGSPATIAL International Workshop on AI for Geographic Knowledge Discovery, 2018 (GeoAI 2018)</i> 2. Zonglin Di, Qi Kang and Mengchu Zhou, “AP-SVM: An Adaptive Clustering-based Support Vector Machine Algorithm for Imbalance Classification”	

	<i>IEEE International Conference on Systems, Man, and Cybernetics, 2018 (SMC 2018)</i>	
	3. Zonglin Di and Qi Kang, “ADPclust-SVM: An Adaptive Clustering-based Support Vector Machine Algorithm for Imbalance Classification” (Abstract Accepted) <i>IEEE International Conference on Networking, Sensing and Control, 2018 (ICNSC 2018)</i>	
	4. Zonglin Di , Xiaoliang Gong, Jingyu Shi, Hosameldin O. A. Ahmed and Asoke K. Nandi, “Detection of Internet Addiction Disorder Based on Personality Questionnaires of Chinese College Students and SVMs” <i>10th International Congress on Image and Signal Processing, BioMedical Engineering and Informatics, 2017 (ICSP-BMEI 2017)</i>	
	5. Xiaoliang Gong, Bozhong Long, Kun Fang, Zonglin Di , Yichu Hou and Lei Cao, “A prediction based on clustering and personality questionnaire data for IGD risk: A preliminary work” <i>12th International Conference on Neural Computation, Fuzzy System and Knowledge Discovery, 2016 (ICNC-FSKD 2016)</i>	
PUBLICATIONS IN PROGRESS	1. Zonglin Di , Xiaoliang Gong, Jingyu Shi, Hosameldin O. A. Ahmed and Asoke K. Nandi, “Internet Addiction Disorder Detection of Chinese College Students Using Several Personality Questionnaire Data and Support Vector Machine” <i>Addictive Behavior Report, Elsevier, 2018</i>	
	2. Zonglin Di , Qi Kang and Mengchu Zhou, “An Adaptive Algorithm for Imbalance Classification Based on Weight-undersampling and Pre-clustering” <i>A transaction paper, 2018</i>	
INTERN EXPERIENCES	Software Intern in National Instruments, Shanghai April 2018 to Presence Framework construction of Auto Test System in Academic Group. Directed by Mr. Dasheng Xu Software Intern in National Instruments, Shanghai July 2017 to Oct 2017 Fix bugs in Diadem data plug-in from customers in German and Japan in Visual Studio 2008 in Windows 7 using C++ Directed by Ms. Mavis Jin Software Intern in Microsoft, Shanghai July 2016 to Sept 2016 ASP.NET WebSite Building and back-end building Directed by Mr. Tarry Wang and Alex Zhang Technical Marketing Intern in Jikexueyuan, Beijing July 2015 to Sep 2015 Solve the Android Problem put forward into the community	
SELECTED AWARDS	• 2nd place in Shanghai College Student Maker Competition Jan 2018 • 3rd prize in Shanghai for National Mathematical Modeling Competition Nov 2017 • Social Scholarship, Tongji University Sept 2017 • 3rd prize in 5th "Taidi" Data Mining Contest Sept 2017 • 3rd prize in 15th Digilent Design Contest July 2017 • 3rd prize in 1st "Huichuangqingchun" Culture & Creativity Project March 2016	
SELECTED PRESENTATIONS	• GeoAI 2018, Seattle, USA Nov 2018 • SMC 2018, Miyazaki, JP Oct 2018 • ICNSC 2018, Hainan, CN March 2018 • ICSP-BMEI 2017, Shanghai, CN Sept 2017 • ICNC-FSKD 2016, Hunan, CN Aug 2016	

- Excellent Undergraduate Sep 2015, 2016, 2017
- LabVIEW Courses for Chinese and International Students Jan 2016 to Sept 2018

SELECTED PROJECTS

Heart Disease Detection using BNN and Diligent PYNQ

3-Layer Convolutional Neural Network implemented in FPGA
Accelerate the speed of training and prediction in a very large scale
Directed by Prof. Wei Zhang

National College Student Innovation Project

Record the action using frame-difference method and upload it to the DropBox in Python and OpenCV on Raspberry Pi
Directed by Prof. Maoheng Sun

5th "Taidi" Data Mining Contest

Nature Language Processing Project using CNN, SVM and KNN
CNN is implemented in TensorFlow and its accuracy is about 89%
Scrapy are used for language material scraping
Directed by Prof. Qi Kang

Shanghai College Student Innovation Project

Intelligent 3D Dressing System with Voice Control Based on Kinect and LabVIEW
Using ASP.NET API and LabVIEW to call the functions of Windows Cortana
A Chinese Patent is in progress Directed by Prof. Zhiming Zhang & Youling Yu

Microsoft-Tongji Innovation Training Project

Participate in the related course organized by Tongji University and Microsoft.
Develop a cross-platform online App using Xamarin and Twilio
Present the result of our group in Microsoft Cooperation, Zizhu, Minhang District on June 17th, 2016.
Directed by Mr. Han Xia

Student Innovation Training Project

Develop a OA System Using C#
Regular Expressions are used in this project
Directed by Ms. Zhihua Wei

SELECTED SOCIAL EXPERIENCE

Virtual Instruments Club

Sept 2015 to Present

Member, Minister of Technical Department, Vice President, President
Cooperate with National Instruments about the related fairs in Tongji University
Response for the daily presentations (200+) to the students for about 20 times
Reference Available from National Instruments Marketing, Great China Area.

Google Developer Group

Dec 2016 to Present

Invited to Google Developer Days 2016 and 2017
Organize the Google DevFes 2017
Organize the Google TensorFlow Technical Salon and take notes for the presenters about the deep learning in computer vision and natural language processing

Unity Unite Shanghai and Beijing

May 2017 and 2018

Keynote and Tech Conference Track Support